

# SERENE BABU

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## CAREER OBJECTIVE

Full Stack Developer with 3 years of experience building modern SPA and API-driven web applications using React, TypeScript, Python, C#, ASP.NET Core, and SQL Server, currently pursuing AWS Developer Associate certification. Experienced in developing REST APIs, secure backend services, reusable frontend architectures, unit testing, Docker containerization, DevOps practices and CI/CD workflows. Seeking full time roles where I can contribute to secure, clean, maintainable software, and cloud-ready deployments.

## KEY COMPETENCIES

- React | NextJs | JavaScript | TypeScript | HTML5 | CSS | Tailwind | Bootstrap | Unit Testing | Integration Testing | UI Testing
- Python Flask | C# | ASP.NET Core (.NET 8) | Entity Framework Core | LINQ | AWS | AI-Assisted Development: Claude Code
- Containerize: Docker | Atlassian Jira | Git & GitHub | DevOps | CI/CD Workflows & Deployment (AWS, Railway)
- Code Reviews | Debugging | Error handling | Problem Solving | SOLID Principles | Clean Architecture

## PROFESSIONAL EXPERIENCE

### Full Stack Developer, *Caddayn, Remote*

Sept 2025 - Present

- Built ASP.NET Core (.NET 8) Web APIs using C#, Entity Framework Core, LINQ, and SQL Server for ERP-based employee management system.
- Designed SQL Server schemas, stored procedures, implemented JWT authentication and role-based authorization.
- Developed REST APIs with pagination, filtering, and centralized error-handling middleware.
- Integrated APIs with the frontend and reduced deployment time by 40% using CI/CD pipelines and Docker containerization.

### Full Stack Developer Volunteer, *Orcawise, Dublin, Ireland*

Dec 2024 – Sept 2025

- Developed a GenAI-powered chatbot platform using React (TypeScript) and Python Flask APIs.
- Built responsive interfaces with dynamic message rendering and asynchronous state handling.
- Integrated OpenAI LLMs (GPT-3.5, GPT-4, GPT-4o) for conversational and structured AI responses.
- Worked with LangChain-based RAG pipelines using embeddings and FAISS vector storage.
- Optimised request flow and improved regeneration response handling logic.

### Front-End Developer, *Springpeople Software Pvt Ltd.*

Nov 2022 - Aug 2023

- Developed a responsive online course platform using HTML5, CSS/SCSS, JavaScript, and jQuery.
- Developed flexible frontend components capable of rendering dynamically managed content within a headless CMS-style architecture.
- Implemented client-side validation, optimized frontend performance for Core Web Vitals compliance.

### Software Development Engineer, *Cloudegic Inc*

Apr 2022 - Nov 2022

- Built React Typescript web applications, unit tested using Jest and React Testing Library with 92% code coverage.
- Implemented centralized theming using styled-components, maintaining consistent design tokens for colors, spacing, and typography across the application.
- Managed complex asynchronous application state using state using hooks, Context API, Redux and Redux-Saga.

## EDUCATION

- **Msc in Computing (Enterprise Software Systems) -2:1 Honours | SETU, Waterford, Ireland | Sept 2023 – Sept 2024**

Thesis – Usability and Accessibility of websites: A case study of applying Design Thinking and WCAG standards

- Developed a React Typescript web application applying WCAG 2.2 and UX best practices.
- Conducted usability research using personas, wireframes, empathy maps, and user journey analysis.
- Tested accessibility compliance using IBM Equal Access Toolkit and Accessibility Insights.

Modules:

Agile software Development – Kotlin, Vue.js, PostgreSQL, Unit tests –JUnit, UI Tests –Selenium, Integration Tests | Enterprise Web Development – React, Typescript, Cloud-native deployment, Automated CI/CD workflows | Cloud Computing – AWS - S3, Lambda, EC2, SNS, SQS, DynamoDB, API Gateway, CloudFront, Cloudformation

- **BTech in Electronics and Communication Engineering | 2017 – 2021**

## AWARDS

- Technical Talk Competition by IEEE – First Position
- Innovation & Entrepreneurship Challenge by IEDC (Innovation and Entrepreneurship Development Cell) SCET - First Place

## PROJECTS

**Spaceplorer** - <https://github.com/serenebabu2022/Spaceplorer>

- Built a full-stack space-data dashboard using React, TypeScript, Node.js, and Express, integrating multiple NASA Open APIs to present astronomy imagery, near-Earth asteroid data, and solar activity.
- Developed interactive data visualisations, asteroid search, date filtering, hazardous-object filtering, and pagination, with responsive UI, loading/error states, and API error handling.
- Implemented unit testing with Jest and React Testing Library across key frontend components and backend functionality to improve reliability and maintainability.
- Currently engineering an AWS-based performance optimisation for API bottlenecks, using **Python-based AWS Lambda functions** for data retrieval/processing and **Amazon ElastiCache** to cache frequently requested NASA data, reducing repeated external API calls and improving response latency